Student Handout — Session 1: Image Capture

What You Will Do in this Session

1. See a live demo of a robot picking objects guided by a camera
2. Learn how to work safely around the robot
3. Capture images of objects that will be used to train an AI model

The Big Picture

```
Camera  →  AI Model  →  "Object is here"  →  Robot picks it
(image)   (segmentation)  (coordinates)        (motion)
```

Today we focus on the **first step**: getting good images for the AI to learn from.

Safety Rules

- **NEVER** put your hands in the robot workspace while it is moving
- Know where the **emergency stop** button is
- Always tell your partner before starting robot motion
- If anything looks wrong, **press the emergency stop**

Your Tasks

Task 1: Test the Camera

1. Open Pylon viewer, connect the Basler camera
2. Activate the camera and capture an image or start a video feed.
3. You should see a live preview from the camera
4. Verify the workspace is visible and well-lit

Task 2: Arrange Objects

1. Place 3–5 geometric shapes on the workspace
2. Make sure they are clearly visible (not blending into the background)

3. Check the camera view — can you see all objects?

Task 3: Capture Images

1. Go to 1. Image Capture section in Jupiter notebook.
2. Go over and run the sections one by one.
3. At "Image Harvesting"; Capture at least **20 images** with different arrangements:
4. [ ] Single object, centered
5. [ ] Single object, off-center
6. [ ] Two objects, separated
7. [ ] Three objects, close together
8. [ ] Five objects, mixed clutter
9. [ ] Objects touching each other
10. [ ] Different orientations (rotated)
11. Continue and run the rest of the sections in Session 1.

Student Handout — Session 2: Annotation and Training

What You Will Do in this Session

1. Upload your images to Roboflow
2. Draw segmentation masks around each object
3. Train an AI model to recognize your objects
4. Evaluate how well the model works

Why Annotation Matters

The AI model learns **only** from the examples you give it. If you label objects sloppily, the model will segment objects sloppily. Your annotations are the ground truth.

Your Tasks

Task 1: Upload Images

1. Log into Roboflow at app.roboflow.com
2. Open your group's project
3. Upload all your good images from Session 1

Task 2: Define Your Classes

Use these exact class names:
- `square`
- `rectangle`
- `circle`
- `triangle`
- `hexagon`
- `ellipse`

Important: Do not add extra classes unless the teacher says so.

Task 3: Annotate

1. Use the "Find objects with AI" when annotating for automatic image detection.
2. Select the class name from the list or enter it before pressing "Find Objects".

3. If it detects to many or to few, modify the confidence threshold.
4. Click save.
5. Repeat this for all images.

Task 4: Train Your Model

1. Go to **Dataset** → **Train Model** -> **Custom Training**
2. Use 10% for validation and 10% for testing.
3. Use the default preprocessing and augmentation settings
4. Click **Train** and select the segmentation model
5. Training takes about 15–20 minutes

Task 6: Record Your API Key

1. Click the settings cog in the left hand side menu -> API keys
2. Go to Jupiter and enter it under **2. Model training (session 2)**
3. Go to Projects in the left hand side menu
4. Select your project.
5. Under the model name it says: ID: /
   Example: markers-uorw8/2
6. Enter the model-id under **2. Model training (session 2)**
7. Enter the version under **2. Model training (session 2)**
8. Run the section and inspect the output:
9. What class was detected?
10. What is the confidence score?
11. Where is the mask in the image?

Student Handout — Session 3: Inference and Calibration

What You Will Do in this Session

1. Run your trained model on images and inspect the results

The Core Problem

In this session we will test the image detection capability.

Your Tasks

Task 1: Setup Inference

1. Go to **3. Inference (session 3)**
2. Run the first section, it just loads the Roboflow project.
3. Run the second section which will register the functions that we will use in the next step.

Task 2: Run Inference on Saved Images

1. Section three will run the inference on a pre-saved image, so run it.
2. Hopefully it will show an image with all parts detected and correctly labled.
3. In the next section you can change the level of inference. If you just run it it will try with 0.2, 0.5 and 0.8 as confidence thresholds.
4. **Try this:** Change the confidence threshold on the line marked `# TODO: adjust threshold`
5. Set it to `0.3` — what happens?
6. Set it to `0.95` — what happens?
7. Choose a good threshold: _

Student Handout — Session 4: Work object calibration

What You Will Do in this Session

1. Calibrate the work object for the robot
2. Draw segmentation masks around each object
3. Train an AI model to recognize your objects
4. Evaluate how well the model works

Why using a work object coordinate fram

The work object (coordinate frame) is used so the robot knows how to translate what the camera sees (ArUco marker pose) into its own motion system. By createing the work object at the same location as the coorcinate system for the ArUco marker, the relation between the camera frame and TCP frame can be linked.

Your Tasks

Task 1: ArUco Calibration

1. Place the ArUco marker flat on the workspace

Task 2: Create a work object

1. Attach the sharp pine to the robot vacium cup, either in the Dobot studio or using the buttoms on the robot.
2. In DOBOT studio, set the robot in online mode
3. In DOBOT studio, select **Settings** -> **User coordiate system**
4. Select the **Camera** coordinate system and press update using Three points settings. If **Camera** does not exist add a new system named **Camera**.
5. Jog or manually drag the robot to the exact bottom left corner/origin of the ArUco marker coordinate system origo and update the first position
6. Jog the robot straight out along the positive X-axis direction and adjust Y position if needed, compare with lines on paper. Update the second position
7. Jog the robot over into the positive Y-axis direction, adjust X position if neede and update the third position. Start from the second X position (unlike ABB)
8. Click Confirm and then Save. The robot calculates the new axes and applies the offset.

TASK 2: Base position

1. Open the Jog menu
2. Jog to the following coordinates in the **Camera** coordinate system.
   X 0
   Y 0
   Z 250
   RX 180
   RY 0
   RZ 90
3. Set the robot in TCP mode.

Student Handout — Session 5: ArUco Calibration

What You Will Do in this Session

1. Learn why the robot needs calibration to understand camera images
2. Perform a simple calibration using an ArUco marker
3. Convert object positions from pixels to millimeters

The Core Problem

The camera sees **pixels** (e.g., "the circle center is at pixel 800, 450").
The robot works in **millimeters** (e.g., "move to 250 mm, 150 mm").

In this session you will figure out the conversion between these two worlds.

Your Tasks

Task 1: Understand the Calibration Concept

Look at the camera image. Answer these questions:
- The image is _ pixels wide and _ pixels tall
- The workspace is approximately _ mm wide (measure with a ruler)
- So roughly, 1 pixel ≈ _ mm (divide workspace width by image width)

This rough estimate shows the idea. Now let's do it precisely.

Task 2: ArUco Calibration

1. **Measure the marker**: _ mm
2. Find the line marked `# TODO: enter your measured marker size` in section 2.
3. Enter your measurement
4. Run the section, it will show you the size you specified.

TASK 3: Base position

1. Open the Jog menu
2. Jog to the following coordinates in the **Camera** coordinate system.
   X 0

Y 0
Z 250
RX 180
RY 0
RZ 90

3. Set the robot in TCP mode.
4. Run the capture section and it will show you an image with the ArUco marker.

Task 4: Verification of your Calibration

1. Place an object on the workspace within the cameras view.
2. Take a ruler and measure the distance from the lower left corner of the ArUco to the center of the marker.
3. At the top of the section find **MANUAL_MEASUREMENT_MM = 130 # <-- CHANGE THIS** enter the correct measurement in mm.
4. Run the verification section to capture an image and find the object's pixel position
5. The program will show an image where it marked the marker and the calculated distance.
6. Is the error between the measured and the calculated acceptable for a suction cup? (hint: ±5–10 mm is usually fine)

Student Handout — Session 6: Pick Point Computation

What You Will Do in this Session

1. Combine inference and calibration to compute pick points in robot coordinates.

Safety Reminder

- Verify emergency stop location.
- Notify the group before pressing "Run"

Your Tasks

Task 1: Verify pick-point

1. In the first section set the height of the pick point over the workspace.
2. A Z value of 0 should be at the same level of the workspace.
3. If the picked items are 2 mm high, set PICK_Z to 2.0.
4. Run the second section and verify the distance from the bottom left of the ArUco to the pick place. in x y z.

Class Conf Pixel (x,y) Work obj (x,y,z) mm

circle 0.92 (855.8, 707.3) (48.8, 75.9, 5.0)

Student Handout — Session 7: Robot Pick and Place

What You Will Do in this Session

1. Connect your vision pipeline to the robot arm
2. Pick objects using camera-guided coordinates

The Full Pipeline (Running Today)

```
Camera captures image
       ↓
Model segments objects
       ↓
Pick point computed (pixel → mm)
       ↓
Robot moves to pick point
       ↓
Suction ON → Lift → Move to drop zone → Suction OFF
```

Safety Reminder

- Emergency stop is at: _____
- **Teacher must approve your first automated pick**
- Tell your partner before pressing "Run"

Your Tasks

Task 1: Robot Configuration

1. The first section contains calibration information.
2. It should be correct so we just have to run it.

Task 2: Connect the Robot

1. Verify in Dobot Studio that the robot is enabled and mode is set to TCP.
2. Run section 2.

3. The output should show **Robot mode 5** and **Robot ready** at the end.

Task 3: Connect the Robot

1. When it is ok run section 3 (step 3a)
2. In section 3b take notice of the VERIFY_Z value, it should NOT be negative. Negative means below the workplane.
3. Run section 3b.
4. Run section 3c and the TCP should point out where the 0.0 coordinate is, i.e. Above the bottom left corner of the ArUco.

Task 4: Capture, Infer, Pick

1. Before running step 4 verify that the setting of all offsets in the jupiter notebook are 0.0.
2. The X_OFFSET = 0.0, etc. represents the tip of the TCP pointer.
3. But we start with 0.0 just to verify that it is working.
4. Run step 5.
5. The robot should:
    - move to above the marker
    - start to suck
    - move to the drop off position
    - stop sucking
    - move to the neutral position
6. Now, adjust the Z_OFFSET so we actually pick up the marker. The Z_OFFSET has to be lowered (into negative numbers) to adjust for using the suction tool without the TCP pointer.
7. You might have to tune the X and Y offsets to make up for measurement errors (lenses).

Student Handout — Session 8: Full pipeline

What You Will Do in this Session

1. Run the complete pipeline

The Full Pipeline

```
Camera captures image
       ↓
Model segments objects
       ↓
Pick point computed (pixel → mm)
       ↓
Robot moves to pick point
       ↓
Suction ON → Lift → Move to drop zone → Suction OFF
```

Safety Reminder

- Emergency stop is at: _____
- Hands **completely clear** of workspace during robot motion.
- Tell your partner before pressing "Run".

Your Tasks

Task 1: Run the Full Pipeline

1. Place **one** object in the workspace
2. Run the full pipeline section

Task 2: Automated Pipeline

1. Go to the next section, **Automated pipeline**
2. Place **several** object in the workspace
3. Find the line **for i in range(1):** And replace **1** with the number of objects.

4. Run the Automated pipeline section

Task 3: Final Demo

1. Set up your best arrangement
2. Demo your system to the teachers
3. Be prepared to explain:
4. How your pipeline works
5. What your success rate is
6. What the biggest source of error was

Task 4: Information About Packing Down the Robot

1. Use the following seetings on the joint values
   J1 0
   J2 125
   J3 - 92
   J4 - 90
   J5 135
   J6 - 4